

reduced power consumption by the fans), low standby power consumption and light-emitting diodes (LEDs) instead of fluorescent lamps are some of the power-saving techniques used in projectors.

LED technology

The last year saw several companies including Acer, Epson, TI and Toshiba introducing models using LEDs. Using LEDs instead of the standard lamps has several advantages. The most notable is their very long life. Rarely does the need arise to change a lamp. Also, the small size of LED lamps offers design flexibility. The ability to switch on or off instantly and deliver better colour saturation, as well as the absence of breakable tubes or filaments, are the other advantages of using LEDs. Not to forget that these lamps do not contain any hazardous mercury or halogen gas, thereby proving to be more environment-friendly.

Mobile projectors

Micro, handheld or pico projectors are gaining momentum. Companies ranging from 3M and Acer to Epson and Optoma have launched pocket projectors recently. These projectors are extremely miniature in terms of the hardware as well as the software in them. Most of them use either a single-LED liquid-crystal on silicon (LCOS) imager chip or TI's DLP technology.

In mid-2009, Nokia spoke about its Pulse projector – a Bluetooth-capable device that combines an LED projector and an NXT speaker with Dolby sound processing. When launched, it will combine beautifully with the Nokia mobile to create a holistic yet easy multimedia experience.

Embedded pico projector technology (such as that offered by TI) is also gaining momentum, and being used to incorporate projectors within mobile devices such as cellphones. Samsung and Epop have already showcased their projector phones, while Nokia and Apple have hinted at the arrival

Some popular brands

Hitachi. Hitachi's ultra-short-throw LCD projector CP-A100 is said to project an image onto a 2m (80-inch) screen from a distance of merely 24.8 cm and hence eliminates the presenter's shadow on the screen during presentations and also reduces projector-glare to a significant level. It has won the *Good Design Award* in Japan for its design. The projector, the company claims, includes a high-quality 3LCD panel to provide crisp, flicker-free images at XGA (extended graphics array) resolution of 1024x768 colour pixels.

NEC. NEC's two new LCD-based multi-function projectors—NP600S and NP500WS—are also claimed to be ultra-short-distance projectors. But the distance they project images from is 1.2 metre and is nowhere close to 24.8 cm of Hitachi CP-A100. What is interesting about the projectors is that they feature panel-crashed air-cooling technology that helps decrease heat outage up to 20 per cent with optical components leading to fewer rotations of the cooling fan and thus lesser power consumption and noise emission. The projectors sport WXGA panel for wide-format images, wide array of input/output terminals and wall-colour correction for vivid images projected directly on uneven surfaces such as walls or blackboards.

LG. LG's HX300G is a compact and lightweight portable projector weighing just 770 gm (1.7 pounds). Featuring 300 lm of brightness with a contrast ratio of 2000:1, this XGA projector uses an LED light source and offers approximately 30,000 hours of lamp life, which is higher than generally offered. Its All File Viewer function, the company says, supports all Microsoft Office files and includes a USB port and support for HD DivX, MP3 and JPEG files, making it ideal for presentations. The projector is expected to hit the market very soon.

ViewSonic. ViewSonic recently came out with a 3D-ready projector called PJD6221—a DLP projector that the company claims to deliver sharp and bright images for virtually any environment. The company says that its BrilliantColor technology produces vibrant colours, while the high brightness and contrast ratio make this projector shine in the classroom or meeting room. It also includes RJ-45 connectivity for remote network control.

Epson. Epson's latest offerings are a series of multimedia projectors, the EB-1910 and EB-1915. The projectors, if the company is to be believed, makes the presentations shine even in broad daylight with brightness level of up to 4000 lm. The EB-1910 also features eight colour mode settings to suit any need, including a new option to combat glare and reflection when projecting on whiteboards.

Acer. Acer recently added to its kitty a 3D-ready DLP projector called XT161. The integrated DLP technology helps the projector actuate microscopic digital mirrors that reflect light for generating a clear picture. High on speed trait, this imaging technology is claimed to produce two images on the screen at the same time—one for the left eye and the other for the right eye—utilising a single projector. Further, 3D glasses conjugate the two images to develop a true 3D effect.

Canon. Canon's XEED range of projectors has been designed for high-end applications such as simulation. The XEED SX800 offers SXGA+ (super extended graphics array plus) resolution. The projector features LCOS (liquid crystal on silicon) technology for producing higher-resolution and higher-contrast images than standard liquid crystal display and plasma display technologies.

of such models in 2010. According to a recent report by research firm In-Stat, the market for embedded pico-projectors will cross \$1.1 billion within five years.

Odds and ends

It is evident that the days of judging a projector merely by obvious parameters such as resolution, brightness and size are over. The current generation of projectors offers many more features, ranging from networking to multi-format support and multiple input/

output interfaces. Moreover, manufacturers offer specific models targeted at certain market segments, such as education or business conferencing. Therefore a well-informed user has more options than he had yesterday, and with a little time and research he can find a model that fits his requirements to the 'T'. ●

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